HARDIK AGARWAL

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Portfolio | LinkedIn | GitHub | LeetCode

Education

Vellore Institute of Technology, Bhopal 2023 – Present
B.Tech in Computer Science Engineering, CGPA: 8.52/10 Expected Graduation: May 2027

St. Joseph's College, Prayagraj 2023
Class XII (ISC): 91.25%

St. Joseph's College, Prayagraj 2021
ICSE Class X: 92.8%

Technical Skills

Languages: Python, Java, SQL, JavaScript
Machine Learning: TensorFlow, Keras, Scikit-learn, CNN, LSTM, GRU, Deep Learning, Time Series Forecasting
Generative AI: RAG, LLMs, Whisper, Ollama, BGE-M3, Semantic Search
Research: Causal Inference, Convergent Cross Mapping (CCM), Neural Granger Causality, Feature Engineering, Model Evaluation
Frameworks & Tools: Next.js, React, Flask, Django, REST APIs, AWS, Git, GitHub
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

Motilal Nehru National Institute of Technology Allahabad May 2026 – Jun 2026
Research Intern (Supervisor: Prof. Anil Kumar Singh) Prayagraj, India
Project Repository

- Reproduced and extended a causality-powered stock forecasting framework using Pearson Correlation, CCM, and Neural Granger Causality on Indian equity markets.
- Built a preprocessing pipeline for 700+ trading days, including chronological splitting, normalization, and sliding-window generation.
- Designed and evaluated DNN, CNN, LSTM, and GRU models using TensorFlow/Keras.
- Evaluated models using RMSE, MAE, MAPE, and R^2, comparing baseline and causal approaches.
- Prepared reproducible experiments and technical documentation.

Projects

RAG-Based Educational Video Assistant *Python – Whisper – Ollama*
GitHub Repository

- Built a Retrieval-Augmented Generation system for semantic search across educational videos.
- Integrated Whisper Large-v2, BGE-M3 embeddings, cosine similarity retrieval, and Llama 3.2 via Ollama.
- Automated transcript generation, vector indexing, and context-aware responses.

repolens – AI-Powered Repository Analysis Tool *Python – AI-GITHUB_repositor-Assistant*

- Developed an AI-powered tool for automated GitHub repository analysis, codebase understanding, and developer productivity.
- Implemented repository structure analysis and code metrics to assess code complexity, duplication, and code quality.
- Integrated LLM-based analysis to generate repository summaries, identify key modules, and suggest refactoring opportunities.
- Designed dependency analysis and visual reports to present repository insights and code health metrics.

Code: GitHub Repository

Certifications

- **Internship Completion Certificate** – MNNIT Allahabad
- **NPTEL Elite Certification** – Machine Learning (IIT Madras)
- Google – Bits and Bytes of Computer Networking
- HackerRank Gold Badges – Java, Python, and C++